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Data Quality

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1. INTRODUCTION

The rapid development of technologies such as Artificial Intelligence (AI), hybrid cloud, Internet of Things (IoT), and Edge computing in recent years has led to the expansive growth of big data sets. This growth has further complicated Master Data Management (MDM) processes, necessitating stricter governance, increased involvement of data stewards, and enhanced data quality assurance measures. Modern data science and machine learning are based on the premise that data is a faithful representation of reality. However, in practice, we frequently encounter the phenomenon where algorithmic complexity cannot compensate for deficiencies within the dataset itself. Understanding data quality is not merely a statistical task, but a critical step in the engineering process that directly determines the success or failure of a predictive model. This chapter will define the core concepts of quality, explore the dimensions through which it is measured, and explain the fundamental principle dictating modern analytics—that the output is only as good as the input.

1.1 Definition and Significance of Data Quality

Data quality is difficult to define as a standalone concept, as data itself can hold different meanings for different users or may be entirely irrelevant to others. Rather, data quality depends on the intended purpose of the dataset; data is considered to be of high quality if it satisfies the requirements of its intended use [1]. Metrics for data quality include accuracy, completeness, consistency, validity, uniqueness, timeliness, interpretability, and others. These measures primarily relate to the state of qualitative or quantitative information within a dataset. Quality assurance is the initial phase of the Knowledge Discovery in Databases (KDD) process, without which all subsequent stages (data mining, evaluation) are compromised. In real-world scenarios, over 60% of the time is dedicated to data verification and preparation, while a smaller percentage is spent on the actual modeling.

Due to the vast volume of data and its integration from multiple disparate sources, modern databases often contain incomplete, imprecise, and inconsistent information. Such information appears in the form of noise, missing values, inconsistencies, or outliers, and can result from either human or computational error. For instance, survey respondents often intentionally enter false data due to privacy concerns, leave optional fields blank, or submit partially incomplete information for various reasons. Data points that exist as legitimate values within a dataset but serve as placeholders are known as "disguised missing data" (DMV). Computational errors may occur, for example, in the form of duplicate tuples. Such poor data quality negatively impacts the reliability and accuracy of results obtained through data mining techniques and machine learning algorithms. Improving data quality is achieved through preprocessing methods such as Data cleaning, Data reduction, Data transformation, and Data selection. While each of these techniques constitutes a specialized field of study and can enhance algorithmic accuracy and efficiency, they will not be discussed in detail in this paper beyond a high-level overview.

1.2 Data Quality Dimensions

The previous subsection listed the data quality dimensions as quality metrics. A description of each follows:

- **Accuracy** – represents the reliability of data and addresses the question: how closely does a data point match its true real-world value? Accurate data is correct, precise, and error-free. For instance, in a column containing years of service, a value of 200 might be found, which was likely entered as 20 by mistake. Statistically, this would be characterized as an **outlier** because 200 is an impossible value for years of service—the data is numerically valid, but inaccurate as it does not reflect reality.
- **Completeness** – addresses the extent to which the data is complete, specifically whether any required data points were not collected. If both a respondent's name and email address were required, but the email is missing, the data is incomplete because contacting the respondent based solely on their name is difficult. A particular challenge here is posed by **DMVs**, where fields are not empty but instead contain generic values (placeholders) that mask the incompleteness.
- **Consistency** – the absence of contradictions within the same dataset or across different data sources. Data is considered consistent if the same information maintains the same value and format throughout the entire system. The simplest example is the method of recording dates within a dataset, where in some instances the format used is DD/MM/YYYY, while in others it is MM/DD/YYYY.
- **Timeliness** – the temporal relevance of data in relation to the moment of its use. In time-sensitive systems, even minor delays of a few seconds (such as a delayed sensor reading) result in low timeliness quality, as the window for decision-making has already passed.
- **Believability** – addresses the extent to which a user trusts the data. This is a subjective rather than a technical metric. A user may question the believability of data based on the source from which it originates or due to observed logical inconsistencies (e.g., a pregnant male).
- **Interpretability** – addresses the extent to which the data is understandable. Undefined column names and encoded values may be meaningless to a user who does not know what they represent. For a machine learning model, different names for the same object represent distinct categories.
- **Validity** – determines the extent to which data adheres to defined rules, formats, and logical constraints specified for the dataset. Examples include whether a grade value falls within the range of 5 to 10, whether the number of digits in a Personal Identification Number (**JMBG**) equals 13 (as required in Serbia), whether the entry format matches the expected format (e.g., date), or if an unexpected data type has been entered.
- **Uniqueness** – ensures that each entity in the dataset appears only once, meaning there are no duplicates. It may occur that a user creates two separate accounts using two different registration methods, thereby causing the same entity to exist within the dataset under two different identifiers.

All the aforementioned quality dimensions are not isolated; they frequently overlap and are mutually dependent. For instance, low validity (incorrect format) often leads to incompleteness (the system cannot parse the data), while compromised uniqueness can create a false representation of the accuracy of certain trends. For a data engineer, these dimensions serve as a checklist that must be completed before data is entrusted to an algorithm.

1.3 Impact of Poor Data Quality on Machine Learning Models

Understanding where and how data fails within these categories allows us to anticipate the problems that will arise during the model training phase, leading us to the fundamental principle of modern analytics—the GIGO (Garbage In, Garbage Out) principle. The GIGO principle dictates that the quality of a model's output is directly dependent on the quality of its input data. Even the most sophisticated algorithms, such as Deep Neural Networks, cannot compensate for inherent deficiencies within a dataset; on the contrary, poor data quality is often amplified through the learning process. For example, the presence of noise and outliers directly violates the assumptions of algorithms such as Linear Regression, where extreme values can drastically shift the regression line and lead to erroneous predictions.

The impact of poor quality manifests through several critical aspects:

- **Model Bias** – Incomplete or non-unique data can lead a model to learn incorrect patterns, favoring certain data groups that are artificially overrepresented.
- **Reduced Accuracy and Reliability** – Inaccurate and inconsistent data increase the error rate, making the model unusable in real-world environments where precision is of vital importance (e.g., medicine or finance).
- **Mathematical Instability** – Issues such as **heteroscedasticity** (unstable variance) or **multicollinearity** (redundancy), which will be discussed in subsequent chapters, directly hinder algorithmic convergence and the interpretation of results.

If data issues such as duplicates, missing values, and outliers are not adequately addressed, organizations increase their risk of negative business outcomes. According to a report by Gartner, poor data quality costs organizations an average of approximately $12.9 million annually [9].

In essence, poor data quality leads to a lack of trust in model performance, as the results become unpredictable and statistically unjustifiable. Therefore, the identification and diagnosis of quality issues do not merely represent a preparatory phase, but rather an essential part of developing a robust and ethically sound machine learning model.

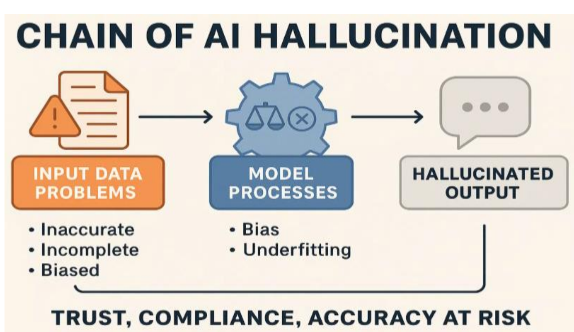

Figure 1. Chain of AI Hallucination [11]

2. Statistical Characterization and Data Profiling

Datasets consist of data objects, more commonly referred to as instances, samples, observations, or entities (among many other terms). Samples are described by synonyms such as: attributes, characteristics, dimensions, or features. More precisely, an attribute represents a property or characteristic of an object that can vary, either from one object to another or over time [2], while a measurement scale is a function that maps a numerical or symbolic value to an object's attribute [2].

Before applying complex machine learning algorithms, it is essential to perform statistical profiling of these attributes. This process allows us to understand the "nature" of the data: where its center lies, how much it varies, and the shape of its distribution. Without such characterization, it is impossible to identify anomalies or make informed decisions regarding data transformation. The first and most crucial step in this process is the identification of attribute types, as the specific mathematical framework and quality metrics to be applied depend entirely on the nature of the attribute itself.

2.1 Attribute Types and Their Impact on Quality Analysis

An attribute type (or data type) can be defined as a set of values an attribute can take, along with a set of operations allowed on those values. In a statistical sense, the attribute type informs us about the nature of the information it carries and defines the mathematical framework within which that attribute can be analyzed. Identifying the type is critical because it sets semantic constraints: the data type suggests whether operations such as addition, ranking, or calculating an average make logical sense. For example, while adding two income values provides clear information, adding two zip codes is mathematically possible but informatively meaningless. From an engineering perspective, the attribute type is the first filter in quality diagnostics, as it dictates the choice of visualization methods (histogram or bar chart) and the statistical tests that will be used in further profiling.

- **Categorical (Qualitative) Attribute Type:**
    - **Nominal** – Attribute values are represented by symbols or names that denote a category, region, or characteristic, and they lack a meaningful order. The term "symbols" does not strictly refer to alphabetic characters; an example of a nominal type is an ID, typically represented by numbers where each ID is unique. This data type serves exclusively to distinguish between samples ($=$ ili $\neq$).
    - **Ordinal** – Attribute values have a logical categorical order, but the differences between these categories are not defined. In other words, the values provide sufficient information only for ordering the samples ($<$ ili $>$). For example, military ranks are clearly ordered from officer to general, but we cannot objectively determine the exact "distance" or difference between them.
    - **Binarny** – Nominal attributes that can have only two possible values: 0 or 1. These can represent presence/absence, true/false, gender, yes/no, etc. Generally, they cover any categorization that can only be described dually. They can be symmetric, if both 0 and 1 are of equal importance (equal weight), or asymmetric, if the specific representation of 1 versus 0 is significant.

- **Numerical (Quantitative) Attribute Type:**
    - **Interval-scaled** – These can be continuous or discrete ordered values where the differences between them are measurable. They can also include negative and zero values (where zero does not denote absence). Differences are defined by units of measurement (e.g., days, millimeters, degrees Celsius).
    - **Ratio-scaled** – Unlike interval-scaled attributes, these possess an absolute zero, and the ratios (proportions) between values are meaningful (e.g., mass, price, number of items sold, temperature in Kelvin).

2.2 Measures of Central Tendency as a Basis for Validation

After identifying the attribute types, the next step in data profiling is the application of measures of central tendency. These measures aim to describe the "center" or the typical value of a data distribution with a single number, providing quick insight into the general characteristics of the sample. However, in the context of quality analysis, measures of central tendency serve as the primary tool for validation and initial anomaly detection. By comparing the mean, median, and mode, an analyst can detect the presence of outliers, distribution asymmetry, or data collection errors. For example, a drastic deviation of the mean from the median is often the first signal that the data contains extreme, inaccurate values that "pull" the average, thereby compromising the future machine learning model. Therefore, these measures do not merely represent descriptive statistics, but rather the foundation upon which the diagnosis of the validity and reliability of the entire dataset is built.

- **Mean** – This refers to the arithmetic mean of a set of values, which is simply calculated using the formula:

$$\bar{x} = \frac{\sum_{i=1}^{N} x_i}{N} \tag{1}$$

  Where N is the number of values, and x_i is the value of the i-th sample.
  The drawback of this measure is its high sensitivity to outliers. If an extremely high or low value appears in the dataset, the mean will shift drastically, providing a false representation of the average. For this reason, a trimmed mean—a mean calculated without considering extreme values—is sometimes used.
- **Median** – The middle value in an ordered set of data that divides the data into upper and lower halves. Unlike the mean, the median is a robust measure, meaning that extreme values do not affect it. For example, if a worker's salary jumps from €1,000 to €100,000 due to a typo, the median will remain the same or change minimally. The downside of this measure is the computational complexity of its calculation when dealing with a large number of samples. An approximation of the median value can be calculated as:

$$median = L_1 + \left(\frac{\frac{N}{2} - (\sum freq)_l}{freq_{median}} \right) width \tag{2}$$

  Where L_1 is the lower bound of the median interval, N the total number of values in the dataset, $(\sum freq)_l$ is the sum of the frequencies of all intervals below the median interval, $freq_{median}$ is the frequency of the median interval, and *width* is the width of the median interval.

- **Mode -** The value that appears most frequently in a dataset. Unlike the previous two measures, the mode can be applied to both numerical and nominal attributes. A dataset can be unimodal, multimodal, or have no mode (if all values appear with equal frequency). The mode is crucial for checking the consistency and validity of categorical data. For example, if we analyze a "City" column and the modes are "Belgrade" and "BELGRADE", we immediately identify a quality issue (inconsistent capitalization). Additionally, in numerical data, an unexpectedly high frequency of a specific value may indicate that the value is being used as a placeholder for missing data. For a unimodal dataset, if the mean and median are known, the mode can be approximated using the formula:

$$mean - mode \approx 3 * (mean - median) \qquad (3)$$

- **Midrange** – The average of the maximum and minimum values in a dataset.

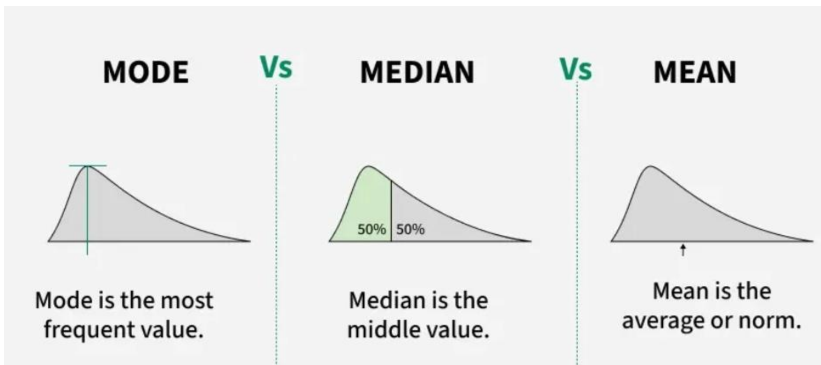

Figure 2. Graphical Representation of the Differences Between Mode, Median, and Mean [12]

2.3 Analysis of Variance and Data Dispersion

Although measures of central tendency provide insight into where data points cluster, they are not sufficient on their own for a complete characterization of a dataset. Two datasets can have an identical mean, yet differ dramatically in terms of their structure and reliability. This is where the analysis of variance and data dispersion (scatter) comes into play. These measures quantify how much samples deviate from their mean, indicating how homogeneous or heterogeneous the dataset is. In data engineering, high dispersion is often an indicator of instability in the data collection process or the presence of significant noise. Understanding variability is of crucial importance for establishing the boundaries of normal data behavior, as it is the measures of dispersion that allow us to mathematically define what constitutes expected variation and what represents an anomaly that compromises the quality of the entire model.

The key measures of data dispersion are:

- **Variance** – The arithmetic mean of the squared deviations of each individual data point from the mean of the entire set. It measures how spread out the data is around the center (the mean), representing the total amount of information in an attribute, and serves as a key indicator of data instability. For N samples, x_1 x_2 ... x_N, where x a numerical attribute and $\bar{x}$ is the sample mean, variance is calculated using the formula:

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2 \tag{4}$$

Since the formula uses the squared deviation$(x_i - \bar{x})^2$, extreme deviations (outliers) have an exponentially greater impact on the final result.

- **Standard Deviation** – Since the numerical value of variance is expressed in the square of the attribute's unit of measurement, standard deviation is more commonly used. It is simply the square root of the varianc:

$$\sigma = \sqrt{\sigma^2} \tag{5}$$

  A small standard deviation indicates that data points are closely clustered around the mean, while a large value indicates that the data is widely dispersed. Considering this measure is crucial for anomaly detection; for example, in data following a normal distribution, values that deviate more than three standard deviations from the mean account for only 0.3% of the samples and are typically treated as outliers. A more detailed analysis of this rule and the normal distribution itself will be the subject of **Chapter 3**. It should be noted that standard deviation is used as a measure of dispersion only if the mean has been previously established as the center of the data.

- **Range, Quartiles, and Interquartile Range** – represents the difference between the maximum and minimum values of an attribute in the dataset. It is another way to spot data irregularities (e.g., Range = 950 for an "Age" column). Quartiles are values that divide an ordered dataset into equal parts. The most common number of divisions is four, in which case each part is called a quartile, representing 25% of the data. If the data is divided into one hundred parts, each is called a percentile. Quartiles indicate the center (50th percentile – the median), spread, and shape of the distribution. The Interquartile Range (IQR) is the difference between the third and first quartiles, measuring the spread of the central 50% of the data. The uniqueness of this measure lies in its complete robustness to outliers, and it can therefore be used to define outlier boundaries as:
  - **Lower bound:** Q1 – 1.5 · IQR
  - **Upper bound:** Q3 + 1.5 · IQR [1]

A particular challenge in variance analysis is the occurrence of **heteroscedasticity**. While **homoscedasticity** implies that the error variance (noise) is constant across the entire dataset, heteroscedasticity occurs when data dispersion changes depending on the value of another attribute. In the context of data quality, this is a critical issue as it indicates that the reliability of our measurements is not uniform. For example, if we analyze household spending, the variance is often much higher among wealthier families than those with lower incomes. The presence of heteroscedasticity directly violates the assumptions of many algorithms, such as linear regression, leading to inaccurate confidence intervals and unreliable predictions from the machine learning model. One method for measuring heteroscedasticity is **Levene's test** [3].

2.4 Distribution Moments: Skewness and Kurtosis

After defining where the data is located (central tendency) and how spread out it is (dispersion), the final step in a complete statistical characterization is analyzing the shape of the distribution. While the mean and variance provide fundamental information, they cannot describe whether a distribution is symmetrical, whether its tails are tilted to one side, or if the center is overly peaked or flat. For these purposes, we use higher-order moments: skewness and kurtosis. In data quality diagnostics, these parameters are crucial because many machine learning algorithms, especially those based on the assumption of normality, lose precision if the data is extremely skewed or has heavy tails (frequent outliers). Understanding the shape of the distribution allows us to make informed decisions regarding necessary data transformations, such as log or Box-Cox transformations, prior to the model training phase.

Skewness can be positive (Right-skewed) – if the tail is on the right side (extending toward the positive part of the x-axis); negative (Left-skewed) – if the tail is on the left side (extending toward the negative part of the x-axis).

Kurtosis indicates the shape of the distribution itself, specifically the height (y-axis value) of the variable relative to its width (x-axis value) [4]. A distribution can be leptokurtotic – peaked; the distribution is too high relative to its tails and has a positive kurtosis value; platykurtotic – flat and wider than a normal distribution, with a negative kurtosis value.

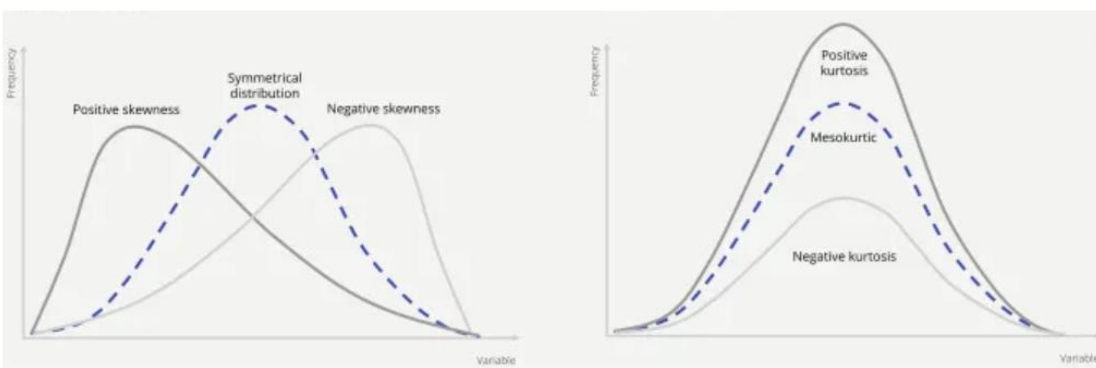

Figure 3. Graphical representations of skewness and kurtosis[8]

3. Distribution Analysis and Data Normality

Distribution analysis represents a key bridge between descriptive statistics and statistical inference. While measures of central tendency and variability describe *what* is happening within the data, distribution analysis tells us *how* those values are distributed across the probability space. At the heart of this analysis lies the normal distribution, which serves as a reference model for quality in data engineering. Understanding the nature of the distribution is essential because it directly dictates the choice of analytical methods: a false assumption of normality can lead to the application of inadequate algorithms, resulting in models that are either overly optimistic or fundamentally flawed. This chapter will focus on diagnostic tools that allow the researcher to determine the degree of alignment between real-world data and theoretical models, while identifying anomalies that may compromise the integrity of future predictions.

3.1 The Importance of Normal Distribution for Statistical Algorithms

The normal distribution, also known as the Gaussian distribution (bell curve), represents a fundamental concept in data analysis, primarily due to its unique role within parametric statistics. Most classical statistical algorithms and tests, such as the **t-test** and **ANOVA**, are based on the assumption that the data follows a normal distribution. On the other hand, non-linear algorithms do not have this assumption, but they typically achieve better performance with such a distribution. The reason for this dominance lies in the predictability of the Gaussian curve, where the mean, median, and mode are identical, and the dispersion strictly obeys the empirical rule of 68.26%-95.44%-99.74%. These figures represent the percentage of values within one, two, or three standard deviations from the center. Thus, in a normal distribution, 99.74% of samples should be within three standard deviations of the center, while anything beyond that is considered a statistically significant extreme value. In the context of data quality, verifying this assumption is critical: if we apply parametric tests to data that significantly deviates from normality, the obtained results (such as p-values and confidence intervals) become unreliable. Therefore, normality analysis is not merely a formal step, but a necessary prerequisite for the validation of any model that relies on the stability of the mean and variance parameters.

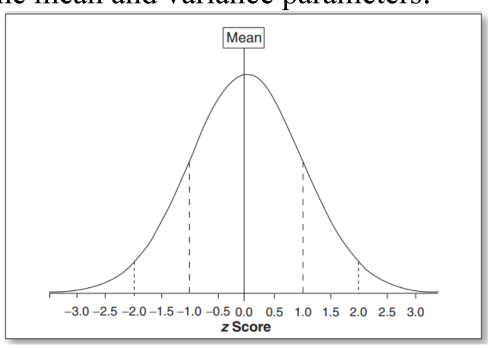

Figure 4. Normal distribution plot [4]

The **Z-score** is used to transform any normal distribution into a standard normal distribution (*mean* = 0 and σ = 1) and is calculated using the following formula:

$$z = \frac{x - \bar{x}}{\sigma} \qquad (6)$$

The t-test is an inferential statistical technique intended to examine the existence of a statistically significant difference between the means of two independent groups. This method is most commonly used in situations where the sample size is relatively small (n < 30), making the assumption of normality critical for its validity. Based on the calculated t-statistic, empirical results are compared against the null hypothesis (which states that the difference between the means is equal to zero) to assess whether the observed differences stem from random variations or a true effect. If the resulting p-value is less than 0.05, the null hypothesis is rejected in favor of the alternative hypothesis. While the t-test is limited to comparing two groups, ANOVA (Analysis of Variance) is used for analyzing three or more groups. By applying the F-test, ANOVA compares the variance within groups and between groups to determine the stability of the observed parameters.

3.2 Visual Diagnostic Methods

Visual diagnostics serve as a preliminary, yet indispensable phase in data quality analysis. Instead of relying on a single numerical value, we use three complementary graphical representations that inform us about the center, spread, and shape of the distribution:

- **Histogram** - divides the range of values into intervals (bins) and displays the number of samples falling into each interval. A histogram allows us to immediately observe skewness, kurtosis, and the distribution itself. If we notice two peaks (bimodal distribution), it signals that the data is likely a mixture of two different processes (e.g., two different sensors) or groups (e.g., male and female animals).
- ***Box-plot*** - based on quartiles, it focuses on data robustness and serves as a diagnostic tool for extreme values. It graphically displays the median and the interquartile range (IQR) within the box, while points located beyond the outer lines (whiskers) represent noise and outliers.
- **Q-Q grafik (*Quantile-Quantile plot*)** – the most precise visual tool, which directly compares the otempirical quantiles of a chosen sample with the quantiles of a theoretical distribution, such as normal or exponential. The plot features a straight line set at a 45-degree angle and points representing the selected data. If the points follow the line, the data is normally distributed; if the ends are curved (S-shape), the data has more extreme values than a normal distribution (higher kurtosis); if they are arched (bow-shaped), the data is skewed (one tail is significantly longer than the other). If the Q-Q plot shows significant deviations at the ends, the results of a t-test or ANOVA become questionable, regardless of what the histogram suggests.

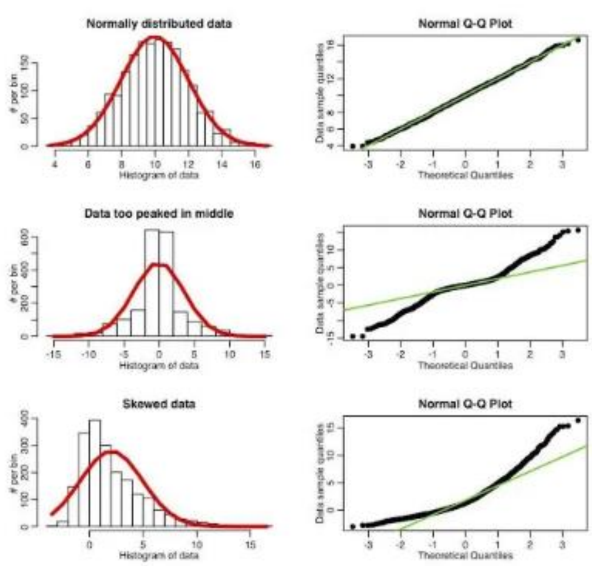

Figure 5. Different distributions as represented on Q-Q plots [13]

3.3 Statistical Normality Tests

Although visual methods provide intuitive and valuable insight into the nature of the data, they always carry a certain degree of subjectivity—what may seem like an acceptable deviation on a Q-Q plot to one analyst might appear as a critical anomaly to another. To reach a final, mathematically grounded decision regarding the fulfillment of the normality assumption, it is necessary to apply formal statistical tests. These tests function within the framework of hypothesis testing, where the null hypothesis (H_0) posits the assumption that the sample originates from a normally distributed population. The test result, expressed through the p-value, provides an exact criterion for rejecting or failing to reject that assumption. The assumption is rejected (non-normal) if $p < 0.05$, and is not rejected (normal) if $p > 0.05$.
The three most commonly used tests are:

- **Sharpiko-Wilk** – analyzes the observed data to assess the degree of symmetry and kurtosis of the distribution. The p-value is calculated by comparing the data to a normal distribution, derived from the sum of squares of observed deviations. It is considered the most powerful test for checking normality, especially for smaller sample sizes. It is highly sensitive to distribution deviations.
- **Kolmogorov-Smirnov** – compares the observed cumulative distribution function (CDF) with the cumulative function of a normal distribution. The p-value is obtained based on the maximum deviation between the two. It is often used for large datasets but is less sensitive than the Shapiro-Wilk test.
- **D'Agostino–Pearson** – first calculates the skewness and kurtosis of the observed distribution and then compares them to a normal distribution to determine the extent of their deviation. The p-value is calculated based on the sum of these deviations. This test can also help identify the specific reason why data is not normally distributed.

It should be noted that these normality tests are primarily conducted to validate the conditions for applying parametric tests. However, normality alone is not sufficient; for the full reliability of these methods, it is also necessary to confirm homoscedasticity and the independence of observations (where the occurrence of one event does not affect the probability of another [3]). Nevertheless, in the domain of machine learning, [3] indicates that algorithm results often fail to meet these strict prerequisites, even when the sample size is theoretically sufficient. Whether dealing with stochastic algorithms whose results vary or deterministic methods with limited samples, deviations from normality are a standard occurrence. In such scenarios, data profiling directly guides us toward the use of non-parametric tests, which are significantly more robust as they do not depend on assumptions regarding the shape of the distribution or the homogeneity of variance.

4. Correlation and Redundancy Analysis

After a detailed examination of the individual characteristics and distributions of each attribute, it is essential to analyze their interrelationships. The quality of a dataset depends not only on the precision of individual measurements but also on the degree of dependency and informativeness of the entire system of variables. Correlation analysis allows us to identify attributes that share similar behavioral patterns, while redundancy analysis helps in recognizing excess information that can unnecessarily burden machine learning models or lead to multicollinearity issues. Utilizing the concepts of similarity and dissimilarity, the goal of this chapter is to establish a framework for selecting the most informative attributes and eliminating those that do not contribute new knowledge to the phenomenon under observation.

4.1 Data Similarity and Dissimilarity Measures

The concepts of **similarity** and **dissimilarity** are fundamental in data profiling because they quantify how close two units of observation are to one another. Informally, according to [2], **similarity (*s*)** is a numerical measure that increases as objects become more alike (usually within the range of [0, 1], where 1 represents total similarity), while dissimilarity **(*d*)** is a measure of distance—the more similar the objects, the smaller the distance. For numerical data, Euclidean distance is most commonly used, whereas for binary or categorical data, measures such as the Simple Matching Coefficient (SMC) or the Jaccard coefficient are applied. Understanding these measures allows us to identify natural groupings within the data or to isolate objects that are so distinct from the rest that they can be considered anomalies.

Minkowski Distance:
$$d(x,y) = \left(\sum_{i=1}^{n} |x_i - y_i|^p\right)^{\frac{1}{p}} \tag{7}$$

For: p = 2 – Euclidean distance, p = 1 – Manhattan distance, and p = ∞ – Supremum distance. It is more frequently used for **dense data**.

Cosine Similarity:
$$\cos(\theta) = \frac{x \cdot y}{\|x\|\|y\|} \tag{8}$$

The value can range from [-1, 1], where 1 indicates that the vectors have maximum similarity. It is more commonly used for **sparse data**.

It is important to note that, to adapt to specific algorithms, it is often necessary to perform a transformation of the proximity measures themselves (e.g., converting similarity to dissimilarity) or to scale them to a standardized interval [0, 1]. This ensures that all relationships within the dataset are mathematically comparable and compatible with the chosen analytical methods. For normalized data, the conversion is as simple as:

$$d = 1 - s \tag{9}$$

Based on this, it can be concluded that a sample whose distance d from other samples is too large (meaning the similarity s is too small) is, in fact, an outlier.

4.2 Correlation Coefficients

While the similarity measures from the previous chapter compare samples (rows), correlation coefficients measure the association between attributes (columns)[1]. For numerical values, the most popular coefficients are:

- **Pearson Correlation Coefficient (*r*)** - measures the strength and direction of the **linear relationship** between two continuous variables:

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{10}$$

  Where n is the number of samples, x_i and y_i are the i-th samples, and $\bar{x}$ and $\bar{y}$ – are the sample means. More generally, the numerator of the formula represents the covariance (which will be discussed later in this paper) of the samples, while the denominator represents the standard deviations of the samples. The value of r can range from [-1,1], where 1 indicates a perfect positive linear relationship, -1 indicates a perfect negative linear relationship, and 0 indicates the absence of a linear relationship. A positive correlation means that as the value of x increases, y also increases; a negative correlation means that as one value increases, the other decreases. The drawback of this measure is its sensitivity to outliers and the prerequisite of a normal distribution of the data. Otherwise, the obtained value may provide a misleading representation of the correlation.

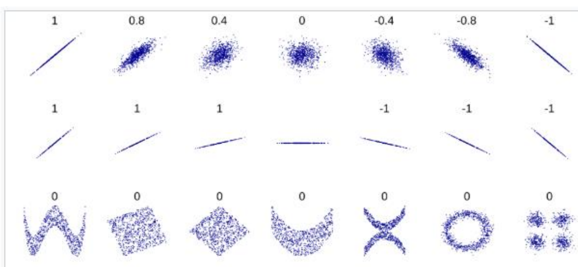

Figure 5. Several sets of (x, y) points and their corresponding correlation coefficients. First row: strength and direction of the linear realtionship; Second row: slope of the relationship; Third row: coefficient values in non-linear cases. [14]

- **Spearman's Rank Correlation Coefficient (*ρ*)** – a non-parametric alternative for measuring monotonic correlation. It does not use the direct values of the data, but rather their **ranks** (position in a sorted list), which makes it significantly more robust to extreme values (outliers). The coefficient's value falls within the same range as Pearson's ([-1, 1]); however, the key difference is that, for example, if the value of *x* increases, the value of *y* does not have to increase linearly—it can, for instance, increase exponentially.

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[1] This is not always the case – correlation can also be used to compare a pair of samples; however, in practice, this is less common due to the diverse attribute types and measurement scales within individual datasets. [2]

- **Chi-square (χ^2) test** – used for categorical data by comparing **observed frequencies** (o) with **expected frequencies** (e) that would exist if the variables were entirely independent. First, a **contingency table** [3] is formed (showing how often two attributes, for example, A and B, occur together), where each cell represents the observed frequency (o_{ij}) of the co-occurrence of values a_i and b_j. The expected frequencies (e_{ij}) are calculated as:

$$e_{ij} = \frac{count(A = a_i) \times count(B = b_j)}{m} \tag{11}$$

Where m is the total number of samples in the dataset, $count(A = a_i)$ is the number of samples with value a_i for attributeA and $count(B = b_j)$ is the number of samples with value b_j for attribute B.

Based on this, χ^2 can be calculated as:

$$\lambda^2 = \sum_{i=1}^{c} \sum_{j=1}^{r} \frac{(o_{ij} - e_{ij})^2}{e_{ij}} \tag{12}$$

Where c and r are the number of distinct categories contained within attributes A and B, respectively. By comparing the obtained result (taking into account the degrees of freedom) with the theoretical distribution, and based on the resulting significance level (p-value), it can be determined whether attributes A and B are statistically associated and redundant.

As an alternative to correlation measures, an exceptionally popular similarity measure—not previously mentioned[2] – is **Mutual Information (MI)**. Based on entropy, it originates from information theory and quantifies how much information about one variable is obtained by observing another. The strength of this measure lies in the fact that it is restricted neither by the type of distribution nor by the nature of the relationship. The normalized value range is [0, 1], where 0 indicates that the variables are entirely independent; the higher the value, the stronger the dependency, regardless of whether it is linear, non-linear, or completely irregular. In practice, these values are typically displayed using a correlation matrix (or dependency matrix) of all attributes in the dataset.

4.3 Redundant Attributes and Multicollinearity

In the data profiling process, high correlation between attributes is often misinterpreted as a positive sign, when it may actually indicate a serious redundancy problem. An attribute is redundant if it provides no new information relative to the set of existing attributes, which most commonly occurs when two variables are directly related (e.g., price with and without tax) or when one variable represents a linear combination of others. A common rule of thumb suggests that a mutual correlation value of **> 0.9** represents redundancy between attributes, allowing one of them to be removed. Naturally, if it can be rationally concluded that these attributes have no logical connection (e.g., geographical location and a physical body characteristic), then there is no need to remove one of them, as they are not redundant attributes but rather distinct separators.

[2] Here, I would like to emphasize the distinction between two terms: while correlation denotes the tracking of mutual data movements (i.e., how they change together), similarity represents a measure of the proximity of data points in a multidimensional space.

A more specific and complex form of this problem is **multicollinearity** — a phenomenon where one independent variable can be predicted with high precision using several other variables simultaneously. The presence of such attributes not only unnecessarily increases the dimensionality of the space and slows down processing, but can also lead to the mathematical instability of the model and complicate the interpretation of individual parameter importance. One measure of multicollinearity is the **VIF (Variance Inflation Factor)**, which measures how much an attribute is associated with all other attributes simultaneously. The VIF is calculated for each attribute individually by setting that attribute as the dependent variable in a regression model, while all other attributes serve as predictors:

$$VIF_i = \frac{1}{1 - R_i^2} \tag{13}$$

Where R_i^2 is the **coefficient of determination** of the regression model for the *i*-th attribute, representing the percentage of that attribute's variance that can be explained by the other variables in the set. For $VIF = 1$ – no correlation exists between the selected attribute and the others (it provides entirely unique information); for $1 < VIF < 5$ – there is a moderate correlation, but it is typically not critical for model stability; for $VIF > 5$ – indicates potentially high (problematic) multicollinearity; $VIF > 10$ – indicates seriously high multicollinearity [15]. A high VIF indicates that the information carried by an attribute is already "distributed" across other attributes, making it a candidate for removal. By eliminating attributes with a high VIF, we reduce model complexity without significant loss of information, thereby directly improving the quality of the dataset.

4.4 Covariance and Variable Interdependence

To complete the discussion on correlation, it is essential to define its foundation—covariance. While correlations are used as standardized measures, covariance provides insight into the very core of a linear relationship. It is a statistical measure that indicates the extent to which two variables change together. If both variables tend to increase simultaneously, the covariance is positive; if one increases while the other decreases, the covariance is negative. The main difference compared to correlation is that covariance is scale-dependent. For example, the covariance between height in meters and weight will be numerically different than if height were expressed in millimeters. This is precisely why correlation (Pearson's coefficient) is defined as normalized covariance (13).

Although correlation is more suitable for interpretation due to its standardization, covariance is indispensable in methods such as Principal Component Analysis (PCA) and Mahalanobis distance. It retains information about the variability (scale) of the data, which allows algorithms to identify which attributes carry the highest intensity of information within the system.

5. Anomaly Detection as a Quality Indicator

Anomaly detection represents one of the most critical aspects of data profiling, as it directly points to potential issues in quality, integrity, or the very process of information generation. In any dataset originating from real-world systems, the presence of values that deviate from expected patterns is inevitable. Identifying these deviations is not merely a technical step toward "cleaning" the data, but a fundamental diagnostic process: it helps us understand whether these values are the result of random measurement errors, systemic sensor failures, or if they represent rare but legitimate events that carry the highest informative value. Anomalies are most commonly manifested as noise, outliers, missing values, artifacts, the presence of samples from foreign distributions, inconsistent data, etc. Each of these anomalies constitutes a distinct and broad research topic in its own right; therefore, this paper will address only the most fundamental principles of defining and detecting these concepts, as well as their overall impact on data quality.

5.1 Taxonomy of Anomalies: Noise and Outliers

Although in a broader context both terms refer to data that deviate from the norm, their origin and treatment in data profiling are fundamentally different. represents a random component of measurement error [2]. It is most often a byproduct of imperfections in measuring instruments, data transmission issues, or the human factor, and is by definition undesirable as it holds no analytical value. However, while noise is filtered out, outliers are investigated in detail because they can reveal new patterns or critical system errors that are not random.

Simply put, an **outlier** is a sample that differs significantly in its value(s) from the other samples in the dataset. If the data distribution is known (any theoretical distribution, e.g., Gaussian), outliers can also be defined as those samples that do not follow the given distribution (the distribution of the majority of the data). As previously mentioned in 3.1, in the case of a normal distribution, all values beyond ±3 standard deviations are considered outliers. Another type of extreme values are the so-called **fringeliers**—samples located at the very edges of the normal distribution (around ±3 σ) which are unlikely to belong to the population of interest, yet still belong to a small extent [4].
Although no universally established classifications exist, common distinctions include class noise and attribute noise [3]. Some of the widely used noise filters include the *Ensemble Filter*, *Cross-Validated Committees Filter*, and *Iterative-Partitioning Filter*, among others [3]. Furthermore, outliers can be categorized into global outliers, contextual (conditional) outliers, and collective outliers [1][3].

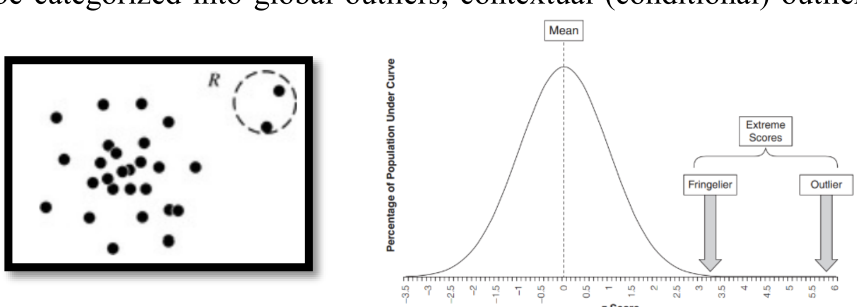

Figure 6. Scatter plot of outliers in a Normal distribution [1] / Outliers and Fringeliers in a normal distribution[4]

[3] These are just a few classifications found in the provided literature that I considered appropriate for this context.

5.2 Statistical Approaches to Outlier Detection

Outlier detection methods vary in their application depending on the dataset being monitored and can be classified based on their mathematical approach as follows[4]:

- **Statistics-based** – These are based on statistical hypothesis testing and fitting the data into specific theoretical distributions. They are divided into parametric methods (which assume a pre-defined distribution, such as Gaussian) and non-parametric methods (e.g., Kernel functions that do not assume a fixed distributional form).
- **Distance-based** – These identify anomalies based on their distance from their nearest neighbors or the density of the space surrounding them. We distinguish between global and local approaches, as well as distance-based versus density-based methods (e.g., the LOF algorithm – *Local Outlier Factor*).
- **Model-based** – These utilize machine learning techniques to build a model of "normal" behavior. These models may assume a single class of normality (e.g., One-Class SVM) or multiple classes of normal states.

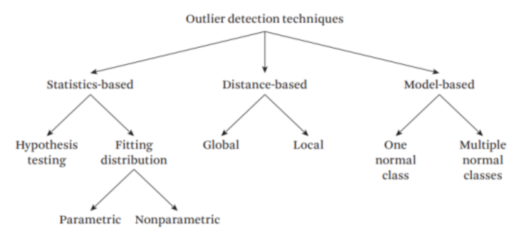

Figure 8.. Classification of Outlier Detection Approaches

As a natural and logical continuation of the data profiling process described in the previous chapters, this paper will focus exclusively on the statistical approach, which directly utilizes distribution parameters and the relationships between attributes to obtain transparent and easily interpretable quality indicators. Unlike complex machine learning models that often function as a black box, statistical methods such as the **Z-score** and **Mahalanobis distance** allow for a precise understanding of *why* a specific data point has been flagged as an anomaly, based on the mathematical structure of the dataset itself.

- **Z-score** – as previously mentioned in 3.1, represents the number of standard deviations by which a specific sample deviates from the arithmetic mean, and is calculated using formula (6). In the data quality profiling process, this method is used for the rapid identification of outliers within individual attributes.
- **Mahalanobis Distance (D_M)** – analyzes a sample relative to the entire multidimensional space, taking into account the covariance between attributes. It practically represents a multidimensional Z-score and is calculated as:

$$D_M(x) = \sqrt{(x-\mu)^T(\sum(x-\mu))^{-1}} \tag{14}$$

[4] The basis for this classification is derived from [6]; for a broader classification, [1] may be consulted.

If the obtained value is high, it is likely an outlier. It is more recommended than the Z-score or Euclidean distance when the attributes are correlated.

5.3 Impact of Extreme Values on Variance and Correlation Measures

The presence of extreme values in a dataset can cause serious deviations in statistical analyses, often leading to erroneous interpretations of data quality by directly increasing unexplained variance and reducing the power of statistical tests by altering the skewness and kurtosis of variables.

5.3.1 Impact on Correlation and Regression

Outliers have a disproportionately large impact on the Pearson correlation coefficient, which is based on the summation of squared deviations from the mean—a single extreme point can create a correlation where none exists or dramatically collapse it in a set where the connection is otherwise strong. In regression analysis, extreme values act as "leverage points" that pull the regression line toward themselves, causing the model to become biased and lose its ability to generalize to new data.

5.3.2 Impact on t-test and ANOVA

As previously described in 3.2, these tests compare the variance between groups with the within-group variance (error). Since outliers drastically increase the within-group variance, they mask actual differences between mean values. This directly leads to a Type II error—the inability to find a statistically significant result that actually exists, simply because the error variance is too large.

5.3.3 Impact on Parameter Estimation

The mean and standard deviation are extremely sensitive to outliers. A single extreme value can shift the average of a set to a point that no longer represents a typical sample, making data profiling inadequate. The most famous example is the change in the average salary in a room when *Bill Gates* walks in [16].

6. Recommendations for Quality Improvement

Following the data profiling process – which includes a detailed analysis of distributions, identification of correlation dependencies, and anomaly detection – the final step in data engineering is defining specific measures for dataset optimization. Profiling results do not serve merely as a static quality report, but as a roadmap for applying adequate transformations and cleaning strategies. Improving data quality at this stage implies a transition from raw, often inconsistent and skewed data, to a structure that is mathematically more stable and aligned with the assumptions of the algorithms to be used in the modeling phase.

6.1 Identifying the Need for Data Transformation

Data transformation is applied when profiling results indicate significant deviations from the assumptions required by machine learning algorithms or statistical tests. The most common indicators for transformation are high skewness, heteroscedasticity (unequal variance), and non-linear relationships between variables.

- **Logarithmic Transformation** – Used to address the issue of positive skewness (Skewness > 1). Taking the logarithm compresses high values and stretches low ones, thereby bringing the distribution closer to normal. It is applied only to strictly positive values and represents a special case of power transformation.
- **Box-Cox** – A family of transformations defined by the parameter χ. It is more sophisticated than the logarithmic transformation because it automatically searches for the optimal parameter that will bring the data closest to a normal distribution. It is applied when the logarithmic transformation does not yield satisfactory results or when an automated process for variance stabilization is desired. If normality tests (such as the Shapiro-Wilk test mentioned in 3.3) show a significant deviation ($p < 0.05$), Box-Cox is used as a universal tool for achieving normality:

$$y^{(\lambda)} = \begin{cases} \dfrac{y^{\lambda} - 1}{\lambda}, & \lambda \neq 0 \\ \ln(y), & \lambda = 0 \end{cases} \tag{15}$$

  Where $y^{(\lambda)}$ is the transformed variable, and χ is the transformation parameter. It is also applied to strictly positive values; however, in cases where negative values are present, an offset parameter can be added. The value of χ is found iteratively by testing different values with small increments within the range of [-3,3] [3].

6.2 Strategies for Treating Identified Noise and Anomalies

Once anomalies have been detected using one of the methods described in **Chapter 5**, it is essential to apply an appropriate treatment strategy. A wrong decision at this stage can lead to the loss of valuable information or, conversely, to leaving behind noise that will compromise future models. The most commonly applied decisions include removal (e.g., when a sample value is illogical relative to the attribute domain), winsorizing (replacing outlier values with boundary values), and imputation (e.g., replacing noise or missing values).

7. Automated Machine Learning (AutoML)

While the previous chapters detail manual and semi-automated profiling methods that require a high degree of engineering expertise, contemporary developments in the field have introduced the concept of **Automated Machine Learning (AutoML)**. AutoML aims to automate the entire model development lifecycle, with a specific focus on the data preprocessing phase, which is often the most demanding task. Instead of an engineer manually identifying the need for a Box-Cox transformation or choosing an outlier treatment strategy, AutoML systems utilize algorithmic optimization to independently test various **transformation pipelines** and select the one that maximizes model performance. However, although these systems dramatically accelerate the process, they do not eliminate the need for understanding the internal data structure; on the contrary, the profiling described in this paper remains a crucial control mechanism, ensuring that automated processes do not become black boxes that produce mathematically optimal but logically unsustainable decisions.

7.1. The Concept of Automated Profiling

The concept of automated profiling is based on shifting the analyst's role to algorithmic systems capable of performing real-time meta-analysis of the dataset. Rather than static observation of histograms or manual calculation of correlation matrices, automated profiling uses heuristics and statistical tests (such as the Shapiro-Wilk test for normality or the VIF factor for collinearity) as input parameters for decision-making. This process begins with automatic data type detection (semantic typing), after which the system applies a set of rules to identify breached integrity, such as missing values or highly skewed distributions. The key contribution of automation lies in its ability to profile Big Data, where manual inspection is physically impossible, generating dynamic reports that automatically rank variables by their predictive power and noise levels. Consequently, the "data discovery" phase is shortened from several days to a few minutes..

7.2. AutoML Tools and Preprocessing Techniques

In practice, the concept of automation is implemented through specific AutoML frameworks that treat data preprocessing not as an isolated step, but as an integral part of an optimization problem. These tools employ various strategies to address data quality issues:

- **Auto-sklearn** – This system utilizes 14 feature preprocessing methods (such as PCA, Kernel PCA, and ICA) and 4 direct data preprocessing methods, including automatic imputation, class balancing, and rescaling. A distinct feature of Auto-sklearn is the use of meta-learning, where the system identifies successful cleaning strategies from similar past problems based on the meta-features of the new dataset.
- ***TPOT* (*Tree-based Pipeline Optimization Tool)*** – Unlike fixed sequences, TPOT uses **genetic programming** to autonomously build and optimize pipelines consisting of a series of feature preprocessors and machine learning models. The goal is to find the most efficient combination of transformations without any user input.
- **Hyperopt-Sklearn and Auto-WEKA** – These early systems introduced an approach where classifier selection and preprocessing module selection are treated as a single, large, joint hyperparameter space (**CASH problem** – *Combined Algorithm Selection and Hyperparameter optimization*). In this way, the system can discover that a specific model (e.g., SVM) performs best with a particular transformation that an engineer might not have manually selected.

- **The Automatic Statistician** – This system goes a step further toward interpretability by generating reports in natural language that explain the preprocessing and modeling steps. In doing so, it bridges the gap between the black box nature of automation and the necessity of understanding decisions made during the data cleaning process [17].

7.3. Industrial Solutions: Cloud AutoML Platforms

In an industrial environment, data profiling and preprocessing have become part of integrated cloud platforms that enable scalability across massive datasets (Big Data). The most prominent representatives of these solutions utilize advanced approaches for data automation:

- **Google Cloud Vertex AI (AutoML):** Google employs technology that relies on deep learning not only for modeling but also for analyzing data structures. Vertex AI automatically recognizes data types and applies sophisticated transformations such as target encoding for categorical variables and automatic rescaling. Particularly significant is their Feature Store, which allows profiled and cleaned features to be reused across different projects, maintaining data integrity at an organizational level.
- **Microsoft Azure AutoML:** Azure stands out for its Data Guardrails system. While the system performs automated preprocessing (such as addressing skewness), it simultaneously generates risk reports. For example, Azure will automatically alert the engineer if it detects excessive skewness, a high number of missing values, or class imbalance, acting as an automated statistical advisor.

- **Amazon SageMaker Autopilot:** Amazon's approach is unique due to its transparency. Although the process is automated, Autopilot generates entire Python notebooks that precisely show how data was profiled, which outliers were removed, and which transformation strategy was applied. This allows engineers to maintain control and auditability over the cleaning process, which is crucial for highly regulated industries such as finance or healthcare.

8. CONCLUSION

Data profiling represents a fundamental process that transforms raw information into a reliable foundation for decision-making and the construction of stable analytical models. Data quality is not an absolute category, but rather the result of precise diagnostics of a dataset's internal structures. By analyzing distributions, determining normality, and identifying skewness, we establish the framework within which statistical tests maintain their validity. Furthermore, by understanding correlations and multicollinearity through the VIF factor, we prevent redundancy that could jeopardize model efficiency, while the application of multivariate methods—such as the Mahalanobis distance—enables the detection of anomalies that remain invisible during standard univariate analyses. Ultimately, it is concluded that profiling is not merely technical preparation but a proactive engineering discipline: identifying the need for transformations like Box-Cox or the appropriate treatment of outliers directly prevents Type II errors and ensures that conclusions derived from data are generalizable and reproducible. In an era dominated by the quantity of information, the ability for in-depth profiling becomes the key separator between algorithms that yield superficial results and those that generate real value based on data integrity.

Throughout this paper, the problem of data impact on models was first identified—poor data quality (noise, outliers, skewness) directly violates algorithmic assumptions. Methods for problem identification were provided—through visualization (Q-Q plots, histograms) and statistical tests (e.g., Shapiro-Wilk). Possible quality assessment solutions were proposed—including distribution analysis (skewness and kurtosis), correlation analysis (e.g., Pearson), and statistical profiling (measures of central tendency and dispersion). The conditions for applying these solutions were defined—depending on the data type (numerical and linear – Pearson), sample size (smaller samples – Shapiro-Wilk; larger samples – Kolmogorov-Smirnov), and the number of outliers. These solutions are grounded in mathematical theory—probability theory, linear algebra, etc. Finally, the pros and cons of these solutions were established—advantages: mathematical proof of quality; disadvantages: the potential loss of the original meaning of transformed features (e.g., the difficulty of explaining the logarithm of a value to a non-technical stakeholder) and the fact that certain results (like the Z-score) are not always representative if the data is not normally distributed.

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